

Sophie Choi

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science, GPA: 3.83

Irvine, CA

Expected June 2028

WORK EXPERIENCE

Software Developer Intern

Jun 2026 – Aug 2026

Naval Air Warfare Center Weapons Division (NAWCWD)

Point Mugu, CA

- Co-authored a bidirectional real-time **C#/.NET** application (**WPF, MVVM**) that parses live radar telemetry from 9 tracked objects into human-readable format and transmits positioning commands all via **UDP** to a \$6M Weibel radar system in real-time, streamlining military range testing operations
- Implemented target tracking visibility during operations by engineering a **multicast networking** integration that sends networking packets to the range's Common Display System application for real-time target visualization
- Presented **system architecture, UML, and XAML GUI** diagrams based on software requirements and **OOP** design principles at critical design reviews, aligning hardware stakeholders on implementation decisions before furthering development
- Authored a comprehensive test suite by writing **unit tests** in **C#**, debugging errors, and producing **application test procedure documentation** to complete **SDLC** validation

EXPERIENCE

Co-Infrastructure Lead

Oct 2025 – Present

Cyber @ UCI Infrastructure Team

Irvine, CA

- Ensured uptime and security for 100+ club members by maintaining a production network built with a **Proxmox** hypervisor cluster (**Linux**), **Docker**, **NAS**, **SSO (Authentik)**, and **VPNs**, continuously improving efficiency through routine maintenance and storage reorganization
- Improved storage bandwidth to 40 GB/s by configuring a dedicated storage switch connecting **Proxmox** hypervisor nodes to a **NAS** using **CLI**-based network configuration
- Deployed high-availability clustering and load balancing with **Docker** containers, eliminating single points of failure to ensure 24/7 access to club infrastructure
- Architected an **Agile**-run sprint project that displays subnets within a /16 network and automates subnet deletion, using **React, Go, RESTful APIs, PostgreSQL, Docker, Kubernetes, GitLab CI/CD, SSO (Authentik), Proxmox**, and **Figma**
- Led **Figma** UI/UX design for a VM provisioning tool serving 20+ internal club members to abstract VM creation, cloning, and deletion, using **React, Go, RESTful APIs, and Docker**

Full-Stack Developer

Oct 2025 – Present

Commit the Change

Irvine, CA

- Programmed a full-stack application using **React, Express.js, PostgreSQL, and CSS**, enabling scalable data management and real-time program oversight for a global NPO operating in 5 continents
- Deployed **RESTful APIs in JavaScript** with **full CRUD functionality** and Promise-based concurrent data fetching, optimizing frontend-backend data flow and improving platform performance by ~25%
- Collaborated in **Agile** sprints with pair programming and **Git** to ship **React** features against 2-week deadlines

Software Developer

Apr 2026 – Present

Blockchain @ UCI Smart Contract Team

Irvine, CA

- Engineered an AI-powered exploit generation engine using **Python, RESTful APIs, and Anvil local chain**, delivering binary proof-of-concept attacks on live contract state rather than theoretical vulnerability reports
- Built a **Python Slither static analysis module** detecting reentrancy, access control, and logic flaws in a smart contract, structuring findings as JSON to feed a downstream **LangGraph agent pipeline**

TECHNICAL SKILLS

Languages: C++, Python, Bash, C#, Go, JavaScript, TypeScript, Java, HTML, CSS, XML

Frameworks/Libraries: React, Node.js, Express, PostgreSQL, .NET, WPF, MVVM, Tailwind CSS

Tools: Docker, Kubernetes, Git, GitLab CI/CD, Proxmox, Terraform, Linux, NAS Administration, Prometheus, Grafana, Figma, Firebase, Unity, Microsoft Office 365

Certifications: Cisco Certified Network Associate (CCNA), Girls Who Code: Data Science for Good